

Title: WATER SUPPLY AND FIELD PURIFICATION METHODS

National Cadet Corps (NCC) â Official Directorate General Training Notes

1. THREATS IN UNTREATED FIELD WATER: Surface water in field terrain contains suspended colloidal silt, organic matter, and dangerous biological pathogens responsible for cholera, amoebic dysentery, typhoid fever, and viral hepatitis A.

2. THE 3-STAGE MILITARY PURIFICATION SEQUENCE: Standard military water treatment follows a sequential 3-step doctrine: Stage 1: Clarification and sedimentation of heavy suspended solids; Stage 2: Physical filtration through clean sand/cloth filters; Stage 3: Chemical sterilization using bleaching powder (calcium hypochlorite) or chlorination tablets (Halazone).

3. BOILING STANDARDS & CHEMICAL DOSING: When chemical reagents are unavailable, water must be brought to a vigorous rolling boil for a minimum of 20 continuous minutes. When using water purification tablets, one tablet treats one liter of clear water, requiring a 30-minute contact time before consumption, maintaining a safe residual chlorine level of 0.5 ppm.

4. STORAGE PROTOCOLS: Purified water must be stored in sterilized, covered galvanised iron (GI) tanks or military water carts with tap dispensers, never accessed via dipping open mugs or hands.